

Project Report – Strategic Intel-Agent

1. Abstract

Strategic Intel-Agent is an AI-powered business intelligence system designed to automate company research and produce professional SWOT analyses. The system integrates real-time web search (Tavily), OpenAI GPT models, LangGraph workflow automation, SQLite state management, and a Streamlit interface. It significantly reduces manual effort and delivers fast, accurate, and structured reports for analysts, students, and decision-makers.

2. Introduction

Market analysis traditionally requires extensive manual effort including searching, reading, and summarizing large amounts of information. This project aims to automate this process through an intelligent multi-agent workflow. The system performs real-time research and generates a professional SWOT analysis, making strategic insights accessible and efficient.

3. Problem Statement

Generating SWOT analysis manually is time-consuming, requires expertise, and involves browsing multiple sources. Many users lack the time or skill to analyze a company thoroughly. Therefore, there is a need for an automated system capable of performing research and generating professional SWOT reports with minimal human intervention.

4. Objectives

- Automate online company research.
- Generate professional SWOT reports using AI.
- Implement a workflow using LangGraph nodes.
- Provide a user-friendly interface via Streamlit.
- Store memory and state using SQLite.
- Integrate email delivery via SMTP.

5. System Architecture

Components:

- Streamlit (Frontend UI)
- Tavily Search API (Real-time research)
- OpenAI GPT (SWOT generation)
- LangGraph (Agent workflow)
- SQLite (Checkpointing)
- SMTP (Email sending)

Workflow:

User → Researcher Node → Writer Node → SWOT Report → Email Delivery

6. Methodology

The system uses two main LangGraph nodes:

Researcher Node:

- Conducts Tavily search.
- Extracts top research insights.

Writer Node:

- Converts collected data into a structured SWOT analysis using GPT.

The UI allows users to review raw research, regenerate searches, and approve final report generation.

7. Key Features

- Automated market research.
- AI-generated SWOT analysis.
- Clean, interactive UI.

- Regenerate research option.
- Email delivery integration.
- Persistent state with SQLite.

8. Results

The system successfully generates complete SWOT reports using real-time web research. Users can view research sources, approve the report, and share it via email. Output is professional, structured, and business-ready.

9. Conclusion

Strategic Intel-Agent automates the end-to-end process of strategic research and SWOT creation. By integrating AI, workflow automation, and real-time search, it offers a powerful tool for analysts, consultants, students, and business teams.

10. Future Enhancements

- Add PDF export functionality.
- Add multi-company comparison.
- Include dashboards and charts.
- Add financial metrics integration.